

Question number	Answer	Additional guidance	Mark
4(a)	whole number in range of 55 to 70 (%) (1)		(1)

Question number	Answer	Additional guidance	Mark
4(b)(i)	A description that includes two of the following points: • {same types of / all (types) / the five types} bacteria (found in gut) in {babies fed with both milks / all babies} (1) • babies fed with breast milk had a {higher proportion of Bifidobacterium and Bacteroides / lower proportion of Enterococcus, Streptococcus and Veillonella} (1) • Bifidobacterium is the highest in babies fed with both types of milk (1)		(2)

Question number	Answer	Additional guidance	Mark
4(b)(ii)	An explanation that includes the following points: • to prevent the growth of pathogenic bacteria / so pathogenic bacteria cannot survive (1) • gut flora compete with other bacteria for {space / nutrients} OR gut flora produce {toxins / lactic acid} (1)		(2)

Question number	Answer	Additional guidance	Mark
4(c)	An explanation that includes three of the following points: - oligosaccharides {provide an alternative to antibiotics that the bacteria are resistant to / can be used on resistant bacteria} (1) - using oligosaccharides removes the selection pressure (1) - oligosaccharides {will have fewer side effects / will not destroy the gut bacteria} (1) - oligosaccharides will increase the gut bacteria which {can then outcompete / will reduce the number of} the {GBS / resistant bacteria} (1)	ACCEPT bacteria are not resistant to oligosaccharides ACCEPT description e.g. reduces the use of antibiotics so number of resistant bacteria does not increase	(3)